



# 3. MAINTENANCE

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## SERVICE INFORMATION

### GENERAL

- Engine oil See page 2-3
- Engine oil filter See page 2-3
- Final drive gear oil See page 2-10

### SPECIFICATIONS

< Engine >

Spark plugs:

| Standard |           | For cold climate (below 5°C, 41°F) |           | For extended high speed riding |           |
|----------|-----------|------------------------------------|-----------|--------------------------------|-----------|
| NGK      | ND        | NGK                                | ND        | NGK                            | ND        |
| DPR8EA-9 | X24EPR-U9 | DPR7EA-9                           | X22EPR-U9 | DPR9EA-9                       | X27EPR-U9 |

Spark plug gap: 0.8—0.9 mm (0.031—0.035 in)

## MAINTENANCE



**HONDA**  
**V45 SABRE**  
**MAGNA**

Ignition timing  
At idle: 10° BTDC  
Advance starts: 10° BTDC at 1,500 rpm  
Full advance: 37° BTDC at 3,300 rpm

Valve clearance  
Cold (Below 35°C, 95°F): Intake/Exhaust: 0.12 mm (0.005 in)

Idle speed: 1,000 ± 100 rpm  
Carburetor synchronization: All carburetors within 60 mm (2.4 in) Hg  
Cylinder compression: 13 ± 2 kg/cm<sup>2</sup> (184 ± 28 psi)  
Throttle grip free play: 2–6 mm (1/8–1/4 in)

### < CHASSIS >

Rear brake pedal free play: 20–30 mm (3/4–1-1/4 in)

Tire:

[ ] ... V45 MAGNA

| Tire size                                          |                                               | Front          | Rear                             |
|----------------------------------------------------|-----------------------------------------------|----------------|----------------------------------|
|                                                    |                                               | 110/90-18 61H  | 130/90-17 68H<br>[130/90-16 67H] |
| Cold tire pressure, psi (kPa, kg/cm <sup>2</sup> ) | Up to 90 kg (200 lbs) load                    | 32 (225, 2.25) | 32 (225, 2.25)                   |
|                                                    | 90 kg (200 lbs) load to vehicle capacity load | 32 (225, 2.25) | 40 (280, 2.80)                   |
| Tire brand                                         | Bridgestone                                   | L303           | G508                             |
|                                                    | Dunlop                                        | F11            | K127C                            |

Suspension air pressure: Front, 6–14 psi (40–100 kPa, 0.4–1.0 kg/cm<sup>2</sup>)  
Rear, 14–57 psi (100–400 kPa, 1.0–4.0 kg/cm<sup>2</sup>)

## TOOLS

Special  
Vacuum gauge set 07404–0020000 or M937B–021–XXXXX (U.S.A. only)  
Carburetor pilot screw wrench 07908–4220201  
Carburetor throttle adjusting bit 07908–MB00000 or KS–AH–008–MBO (U.S.A. only)  
Common Adjusting wrench 07708–0030200 or Snap-on Tool Wrench-X0810  
or equivalent available in U.S.A.



## MAINTENANCE SCHEDULES

### V45 SABRE-VF750S

Perform the PRE-RIDE INSPECTION in the Owner's Manual at each scheduled maintenance period.

I: INSPECT AND CLEAN, ADJUST, LUBRICATE, OR REPLACE IF NECESSARY.

C: CLEAN, R: REPLACE, A: ADJUST, L: LUBRICATE

| FREQUENCY                  |                                      | WHICHEVER<br>COMES<br>FIRST<br>↓ | ODOMETER READING (NOTE 3) |                        |                         |                          |                          |                          |                          | Refer to page |
|----------------------------|--------------------------------------|----------------------------------|---------------------------|------------------------|-------------------------|--------------------------|--------------------------|--------------------------|--------------------------|---------------|
|                            |                                      |                                  | 600 mi<br>(1,000 km)      | 4,000 mi<br>(6,400 km) | 8,000 mi<br>(12,800 km) | 12,000 mi<br>(19,200 km) | 16,000 mi<br>(25,600 km) | 20,000 mi<br>(32,000 km) | 24,000 mi<br>(38,400 km) |               |
| EMISSION RELATED ITEMS     | ITEM                                 | EVERY                            |                           |                        |                         |                          |                          |                          |                          |               |
|                            | * FUEL LINES                         |                                  |                           |                        | I                       |                          | I                        |                          | I                        | 3-5           |
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|                            | CRANKCASE BREATHER                   | NOTE 2                           |                           | C                      | C                       | C                        | C                        | C                        | C                        | 3-9           |
|                            | SPARK PLUGS                          |                                  |                           | R                      | R                       | R                        | R                        | R                        | R                        | 3-9           |
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|                            | * CARBURETOR-SYNCHRONIZE             |                                  | I                         |                        | I                       |                          | I                        |                          | I                        | 3-12          |
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|                            | * FINAL DRIVEN FLANGE                | TIRE CHANGE                      |                           |                        | L                       |                          | L                        |                          | L                        | 17-7          |
|                            | BATTERY                              | MONTH                            | I                         | I                      | I                       | I                        | I                        | I                        | I                        | 3-15          |
|                            | BRAKE FLUID (FRONT)                  | MONTH 1<br>2 YEARS* R            | I                         | I                      | I                       | I                        | I                        | I                        | *R                       | 3-15          |
|                            | BRAKE SHOE/PAD WEAR                  |                                  |                           | I                      | I                       | I                        | I                        | I                        | I                        | 3-16          |
|                            | BRAKE SYSTEM                         |                                  | I                         |                        | I                       |                          | I                        |                          | I                        | 3-16          |
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|                            | CLUTCH FLUID                         | MONTH 1<br>2 YEARS* R            | I                         | I                      | I                       | I                        | I                        | I                        | *R                       | 3-18          |
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\* SHOULD BE SERVICED BY AN AUTHORIZED HONDA DEALER, UNLESS THE OWNER HAS PROPER TOOLS AND SERVICE DATA AND IS MECHANICALLY QUALIFIED.

\*\* IN THE INTEREST OF SAFETY, WE RECOMMEND THESE ITEMS BE SERVICED ONLY BY AN AUTHORIZED HONDA DEALER.

NOTES: 1. SERVICE MORE FREQUENTLY WHEN RIDING IN DUSTY AREAS.

2. SERVICE MORE FREQUENTLY WHEN RIDING IN RAIN OR AT FULL THROTTLE (U.S.A. ONLY).

3. FOR HIGHER ODOMETER READINGS, REPEAT AT THE FREQUENCY INTERVAL ESTABLISHED HERE.





# V45 MAGNA-VF750C

Perform the PRE-RIDE INSPECTION in the Owner's Manual at each scheduled maintenance period.

I: INSPECT AND CLEAN, ADJUST, LUBRICATE, OR REPLACE IF NECESSARY.

C: CLEAN, R: REPLACE, A: ADJUST, L: LUBRICATE

| FREQUENCY                  |                                      | WHICHEVER<br>COMES<br>FIRST<br>↓ | ODOMETER READING (NOTE 3) |                        |                         |                          |                          |                          |                          |      | Refer to page |
|----------------------------|--------------------------------------|----------------------------------|---------------------------|------------------------|-------------------------|--------------------------|--------------------------|--------------------------|--------------------------|------|---------------|
|                            |                                      |                                  | 600 mi<br>(1,000 km)      | 4,000 mi<br>(6,400 km) | 8,000 mi<br>(12,800 km) | 12,000 mi<br>(19,200 km) | 16,000 mi<br>(25,600 km) | 20,000 mi<br>(32,000 km) | 24,000 mi<br>(38,400 km) |      |               |
| ITEM                       |                                      | EVERY                            |                           |                        |                         |                          |                          |                          |                          |      |               |
| EMISSION RELATED ITEMS     | * FUEL LINES                         |                                  |                           |                        | I                       |                          | I                        |                          | I                        | 3-5  |               |
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|                            | CRANKCASE BREATHER                   | NOTE 2                           |                           | C                      | C                       | C                        | C                        | C                        | C                        | 3-9  |               |
|                            | SPARK PLUGS                          |                                  |                           | R                      | R                       | R                        | R                        | R                        | R                        | 3-9  |               |
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|                            | ENGINE OIL                           | YEAR                             | R                         |                        | R                       |                          | R                        |                          | R                        | 2-3  |               |
|                            | ENGINE OIL FILTER                    | YEAR                             | R                         |                        | R                       |                          | R                        |                          | R                        | 2-3  |               |
|                            | * CARBURETOR-SYNCHRONIZE             |                                  | I                         |                        | I                       |                          | I                        |                          | I                        | 3-12 |               |
|                            | * CARBURETOR-IDLE SPEED              |                                  | I                         | I                      | I                       | I                        | I                        | I                        | I                        | 3-13 |               |
|                            | RADIATOR COOLANT                     |                                  |                           |                        | I                       |                          | I                        |                          | *R                       | 3-13 |               |
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| NON-EMISSION RELATED ITEMS | * COOLING SYSTEM HOSES & CONNECTIONS |                                  | I                         |                        | I                       |                          | I                        |                          | I                        | 3-14 |               |
|                            | FINAL DRIVE OIL                      |                                  |                           |                        | I                       |                          | I                        |                          | R                        | 2-10 |               |
|                            | * FINAL DRIVEN FLANGE                | TIRE CHANGE                      |                           |                        | L                       |                          | L                        |                          | L                        | 17-7 |               |
|                            | BATTERY                              | MONTH                            | I                         | I                      | I                       | I                        | I                        | I                        | I                        | 3-15 |               |
|                            | BRAKE FLUID (FRONT)                  | MONTH I<br>2 YEARS* R            | I                         | I                      | I                       | I                        | I                        | I                        | *R                       | 3-15 |               |
|                            | BRAKE SHOE/PAD WEAR                  |                                  |                           | I                      | I                       | I                        | I                        | I                        | I                        | 3-16 |               |
|                            | BRAKE SYSTEM                         |                                  | I                         |                        | I                       |                          | I                        |                          | I                        | 3-16 |               |
|                            | * BRAKE LIGHT SWITCH                 |                                  | I                         |                        | I                       |                          | I                        |                          | I                        | 3-17 |               |
|                            | * HEADLIGHT AIM                      |                                  | I                         |                        | I                       |                          | I                        |                          | I                        | 3-17 |               |
|                            | CLUTCH FLUID                         | MONTH I<br>2 YEARS* R            | I                         | I                      | I                       | I                        | I                        | I                        | *R                       | 3-18 |               |
|                            | CLUTCH SYSTEM                        |                                  | I                         |                        | I                       |                          | I                        |                          | I                        | 3-18 |               |
|                            | SIDE STAND                           |                                  |                           |                        | I                       |                          | I                        |                          | I                        | 3-18 |               |
|                            | * SUSPENSION                         |                                  | I                         |                        | I                       |                          | I                        |                          | I                        | 3-19 |               |
|                            | * NUTS, BOLTS, FASTENERS             |                                  | I                         |                        | I                       |                          | I                        |                          | I                        | 3-21 |               |
|                            | ** WHEELS                            |                                  | I                         |                        | I                       |                          | I                        |                          | I                        | 3-21 |               |
|                            | ** STEERING HEAD BEARINGS            |                                  | I                         |                        | I                       |                          | I                        |                          | I                        | 3-21 |               |

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NOTES: 1. SERVICE MORE FREQUENTLY WHEN RIDING IN DUSTY AREAS.

2. SERVICE MORE FREQUENTLY WHEN RIDING IN RAIN OR AT FULL THROTTLE (U.S.A. ONLY).

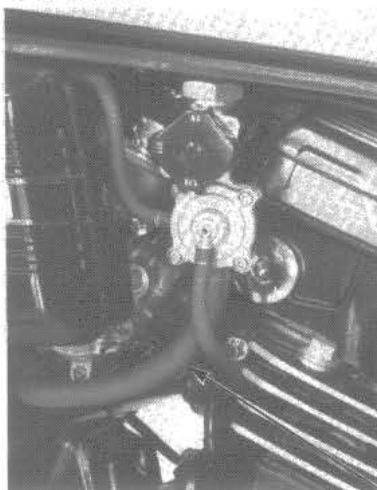
3. FOR HIGHER ODOMETER READINGS, REPEAT AT THE FREQUENCY INTERVAL ESTABLISHED HERE.



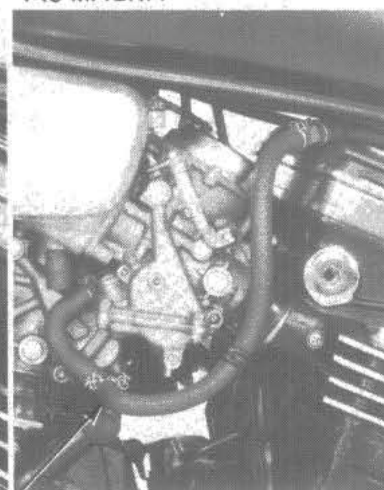
## FUEL LINES

Replace any parts which show deterioration, damage or leakage.

V45 SABRE



V45 MAGNA



FUEL LINES

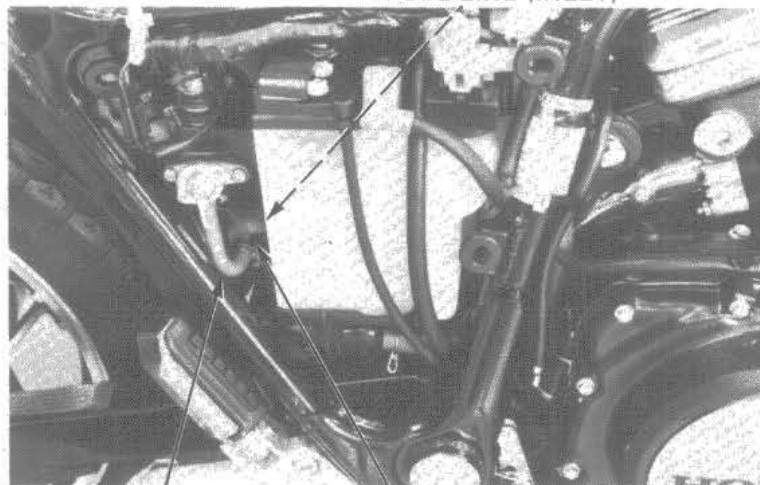
## FUEL FILTER(V45 MAGNA)

### WARNING

*Gasoline is flammable and is explosive under certain conditions. Do not smoke or allow flames or sparks in your working area.*

Turn the fuel valve OFF.  
Remove the right side cover.  
Remove the fuel filter from the holder.  
Clip the fuel line (inlet side) closed.  
Disconnect the fuel lines from the filter.  
Replace the fuel filter with a new one when indicated by the maintenance schedule (See page 3-4).  
Install the fuel filter and reconnect the fuel lines.  
After installing, turn the fuel valve ON and check that there are no fuel leaks.

FUEL LINE (INLET)



FUEL LINE  
(OUTLET)

FUEL FILTER





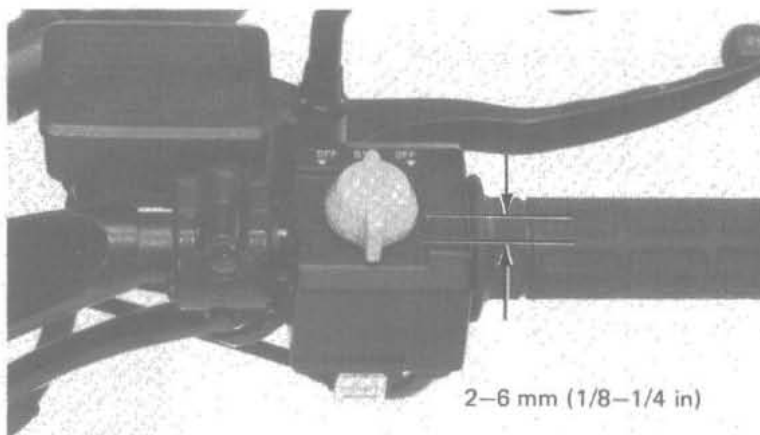
## THROTTLE OPERATION

Check for smooth throttle grip full opening and automatic full closing in all steering positions. Check the throttle cables and replace them if they are deteriorated, kinked or damaged.

Lubricate the throttle cables (page 2-11), if throttle operation is not smooth.

Measure throttle grip free play at the throttle grip flange.

**FREE PLAY: 2–6 mm (1/8–1/4 in)**

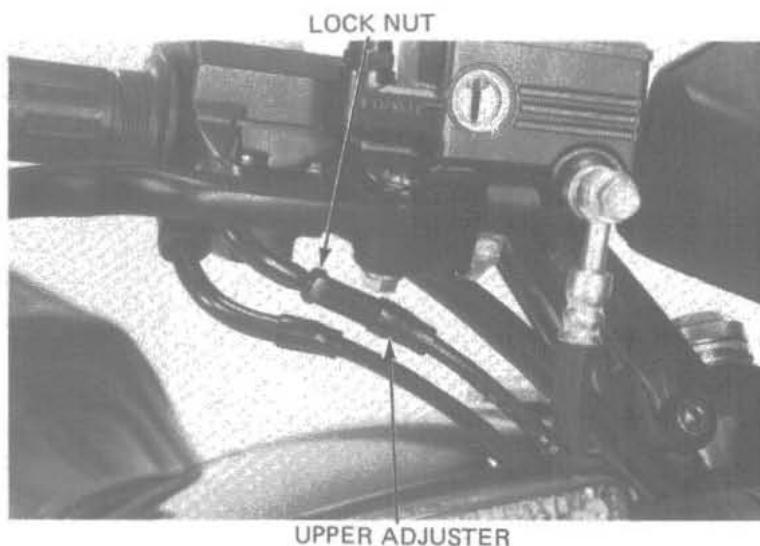


Adjustment can be made at either end of the throttle cable. Minor adjustments are made with the upper adjuster and major adjustments are made with the lower adjuster after removing the fuel tank (V45 SABRE) or lifting the fuel tank up (V45 MAGNA).

Adjust by loosening the lock nut and turning the adjuster.

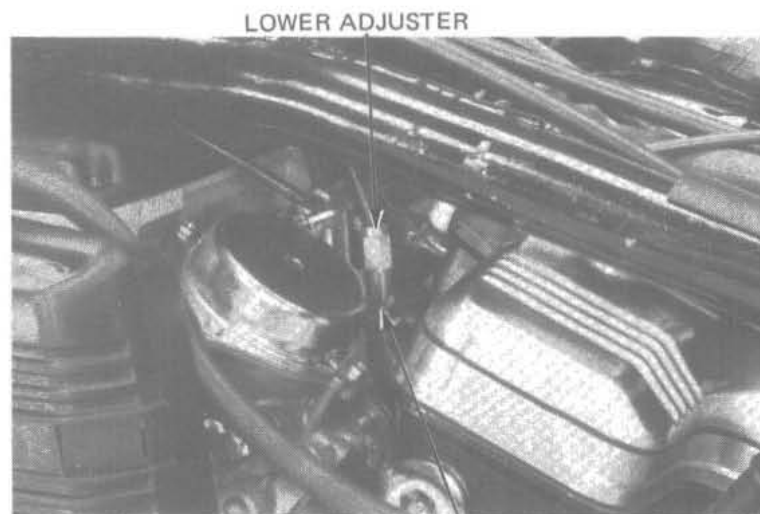
Tighten the lock nut and recheck throttle operation.

Install the fuel tank, if removed.



LOCK NUT

UPPER ADJUSTER



LOWER ADJUSTER

LOCK NUT



## CARBURETOR CHOKE

The V45 choke system uses a fuel enriching circuit controlled by a bystarter valve. The bystarter valve opens the enriching circuit via cable when the choke lever on the handlebar is pushed up. Check for smooth upper choke lever operation. Lubricate the choke cable, if the operation is not smooth.

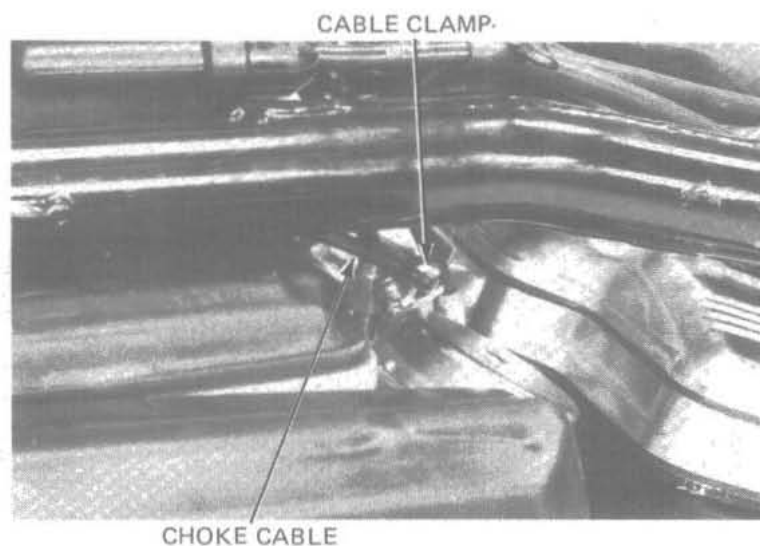
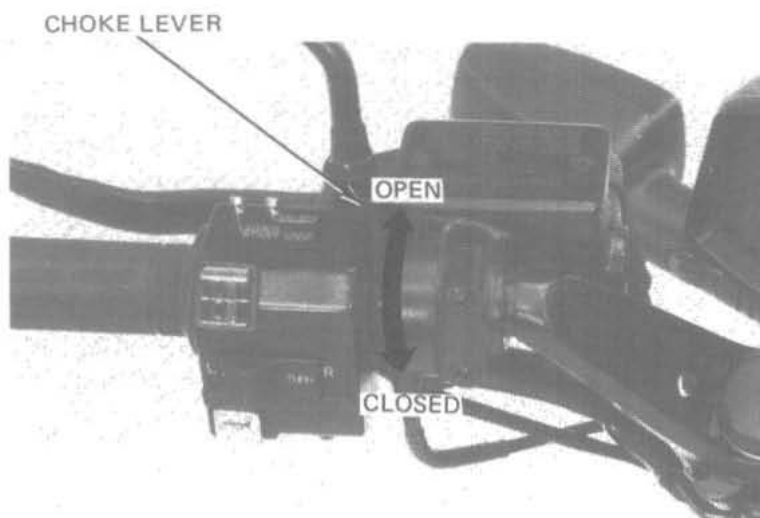
Push the choke lever on the handlebar all the way up to fully open. Make sure the choke valve is open by trying to move the choke lever on the carburetor, after removing the fuel tank (V45 SABRE) or lifting the fuel tank up (V45 MAGNA). There should be no free play.

Adjust if necessary, by loosening the choke cable clamp on the carburetor and moving the choke cable casing so the choke lever is fully open.

Tighten the clamp.

Pull the choke lever all the way to fully closed. Make sure the choke valve is fully closed by checking for free play in the cable between the lever on the carburetor and cable casing.

Reinstall the removed parts in the reverse order of disassembly.



## AIR CLEANER

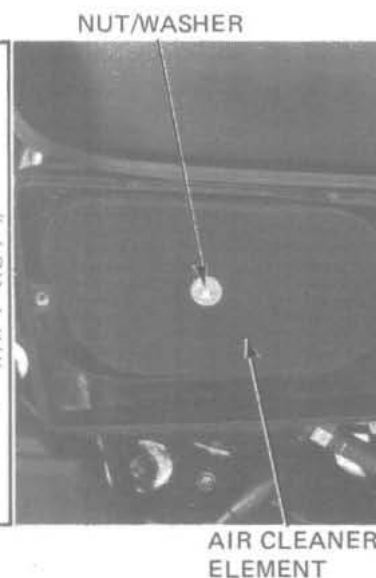
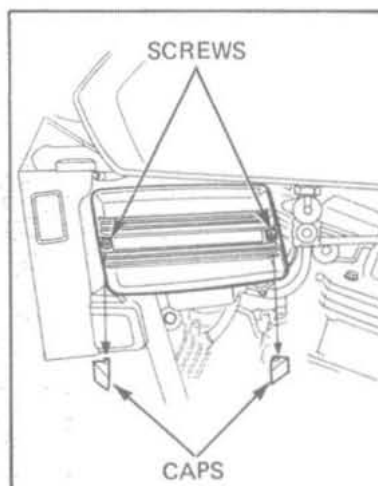
### V45 SABRE:

Remove the caps from the left and right air cleaner cover screws.

Remove the air cleaner cover screws and the two covers.

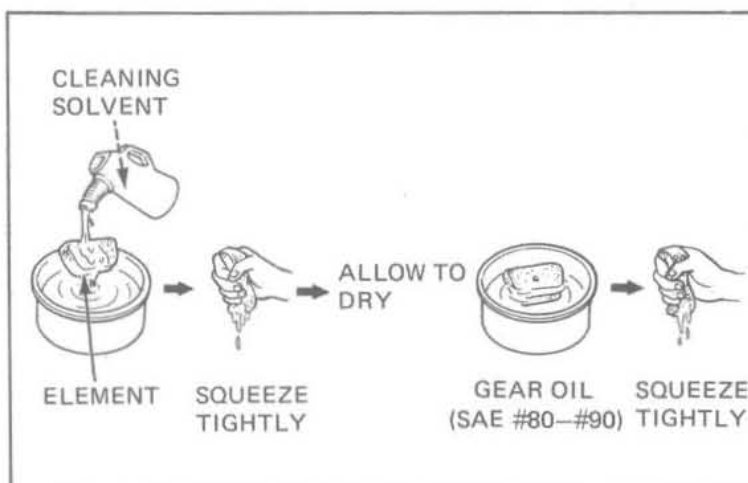
Remove the nut and washers and the two air cleaner elements.

Remove the elements from their holders.



Wash the elements in non-flammable or high flash point solvent. Squeeze out and let them dry.  
Soak the elements in gear oil (SAE #80-#90) and squeeze out the excess.

Install the removed parts in the reverse order of disassembly.



## V45 MAGNA:

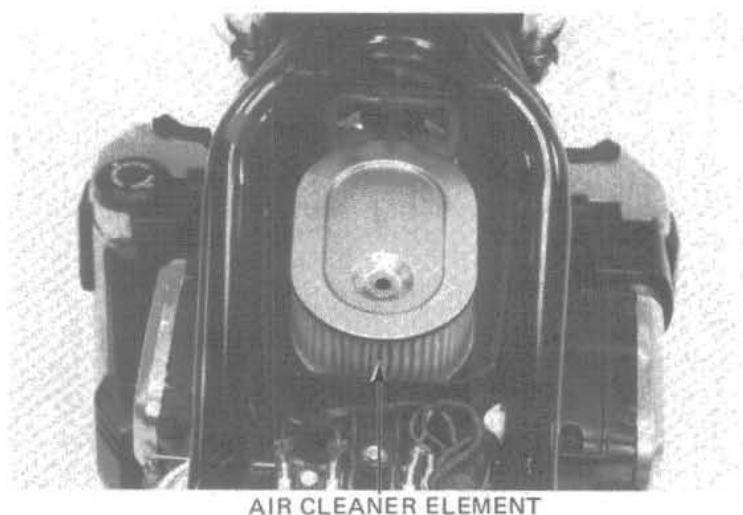
Make sure the fuel tank is less than half-full. Drain it, if it is more than half-full.

Remove the tank mounting bolts and lift the fuel tank up and hold it in place with the tank support rod.

Remove the air cleaner cover screws.



Remove the air cleaner element and discard it in accordance with the maintenance schedule.

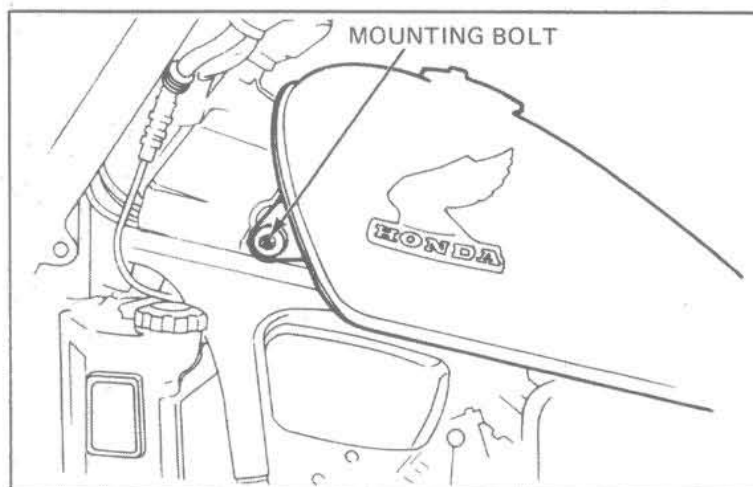






Also, replace the element if it is excessively dirty, torn or damaged.

Install a new element and install the cover.  
Disengage the tank support rod by lifting up slightly.  
Let the fuel tank drop into place and install the fuel tank mounting bolts. Be careful not to pinch or kink the vent and fuel lines.



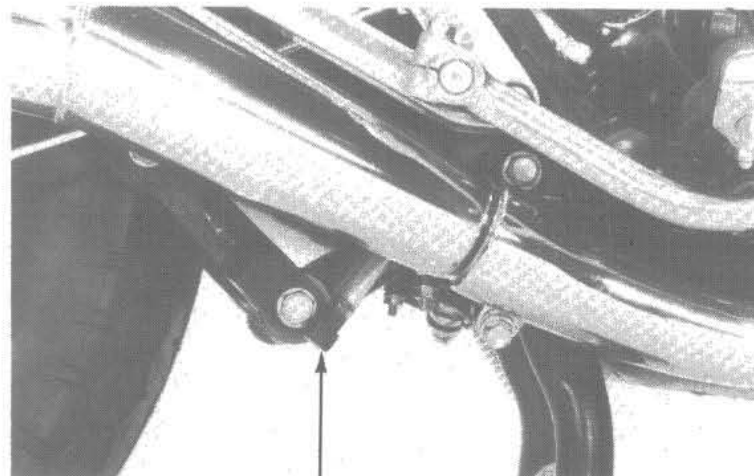
## CRANKCASE BREATHER

Remove the plug from the drain tube to empty any deposits.

Install the drain plug.

### NOTE:

Service more frequently when riding in rain or at full throttle, or if the deposit level can be seen in the transparent section of the drain tube.



## SPARK PLUGS

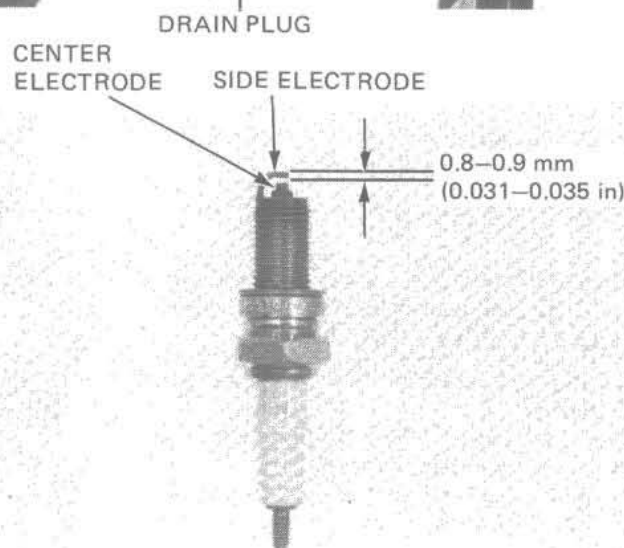
### RECOMMENDED SPARK PLUGS

|                                       | NGK      | ND        |
|---------------------------------------|----------|-----------|
| Standard                              | DPR8EA-9 | X24EPR-U9 |
| For cold climate<br>(Below 5°C, 41°F) | DPR7EA-9 | X22EPR-U9 |
| For extended<br>high speed riding     | DPR9EA-9 | X27EPR-U9 |

Disconnect the spark plug caps.  
Clean any dirt from around the spark plug bases.  
Remove and discard the spark plugs.  
Measure the new spark plug gaps using a wire-type feeler gauge.

**SPARK PLUG GAP: 0.8–0.9 mm (0.031–0.035 in)**

Adjust by bending the side electrode carefully.  
With the plug washer attached, thread each spark plug in by hand to prevent crossthreading.  
Tighten the spark plugs another 1/2 turn with a spark plug wrench to compress the plug washer.  
Connect the spark plug caps.





## IGNITION SYSTEM

### NOTE:

The ignition system is transistorized and cannot be adjusted. If the ignition timing is incorrect, check the spark unit and pulse generator and replace any faulty parts (Section 20).

Remove the alternator and left crankcase covers.  
 Wipe off the oil from the flywheel.  
 Connect the timing light to the high tension wire of the No. 1 cylinder.  
 Start the engine and let it idle.

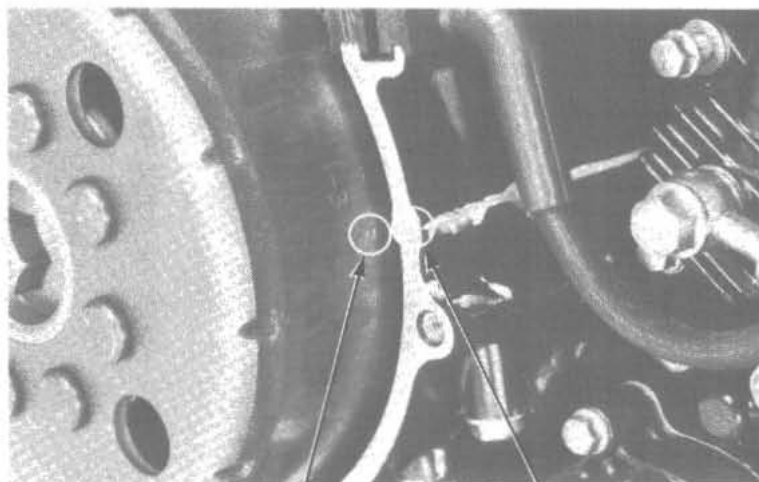
**IDLE SPEED: 1,000 ± 100 rpm**

The timing is correct if the "F" mark on the flywheel aligns with the crankcase rear mating surfaces.

ALTERNATOR COVER



LEFT CRANKCASE COVER



"F" MARK

REAR CRANKCASE  
MATING SURFACES

## VALVE CLEARANCE

### NOTE

Inspect and adjust valve clearance while the engine is cold. (Below 35°C, 95°F)

Drain coolant from engine and frame (page 6-3).

### NOTE

Drain the coolant into a clean container for reuse. Scheduled coolant replacement is every 2 years or 24,000 miles (38,600 km).

Remove the right and left frame side covers and remove the seat.

Turn the fuel valve "OFF" (V45 MAGNA) and remove the fuel tank (V45 SABRE).

Lift up the fuel tank by removing the front mounting bolts (V45 MAGNA). Be sure the tank is less than half full.

If not, drain the tank.

Remove the spark plug caps.



SIDE COVER





Remove the radiator (See page 6-8).

Remove the front and rear cylinder head cover bolts and both cylinder head covers.

Remove the alternator and left crankcase covers.

## INSPECTION

Measure and adjust the intake and exhaust valve clearances as described below.

Rotate the crankshaft clockwise to align the T1.3 mark with the crankcase mating surfaces.

Make sure the No. 1 piston is at TDC (Top Dead Center) on the compression stroke.

Check the valve clearances for the No. 1 cylinder using two feeler gauges for each pair of valves; one for each valve that shares a common rocker arm.

### VALVE CLEARANCE (IN, EX):

0.12 mm (0.005 in)

If adjustment is needed, loosen the lock nuts and turn the adjusting screws until there is a slight drag on both feeler gauges. Both feeler gauges should remain inserted during adjustment.

Hold the adjusting screws and tighten the lock nuts.

**TORQUE: 21–25 N·m (2.1–2.5 kg-m, 15–18 ft-lb)**

### CAUTION:

*The lock nuts will come loose if not tightened to the correct torque value.*

Recheck the valve clearance.

Rotate the crankshaft 270° clockwise to align the T2.4 mark with the crankcase mating surfaces and check the valve clearances for the No. 2 cylinder.

Adjust using the procedures for the No. 1 cylinder.

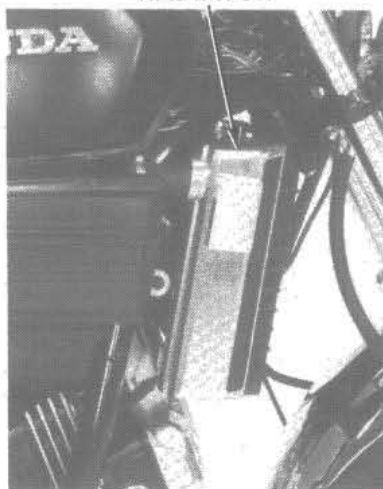
Rotate the crankshaft 90° clockwise to align the T1.3 mark with the crankcase mating surfaces and check the valve clearances for the No. 3 cylinder.

Adjust using the procedures for the No. 1 cylinder.

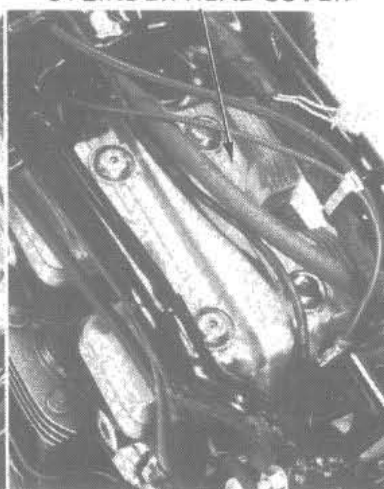
Rotate the crankshaft 270° clockwise to align the T2.4 mark with the crankcase mating surfaces and check the valve clearances for the No. 4 cylinder.

Adjust using the same procedures as for the No. 1 cylinder.

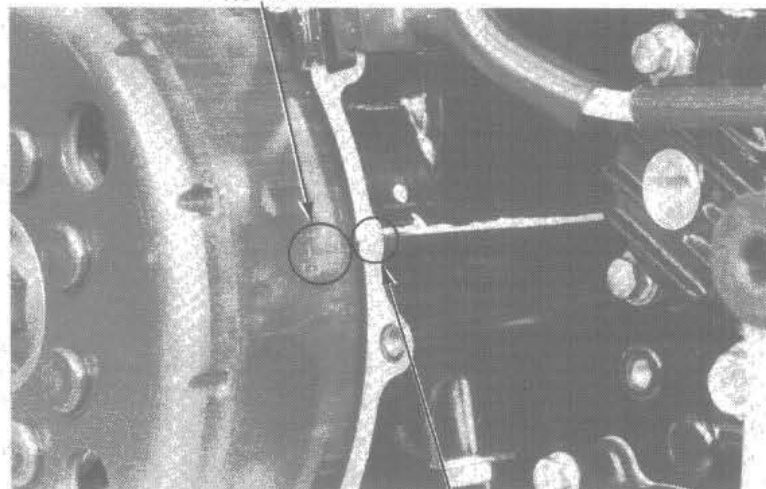
RADIATOR



CYLINDER HEAD COVER

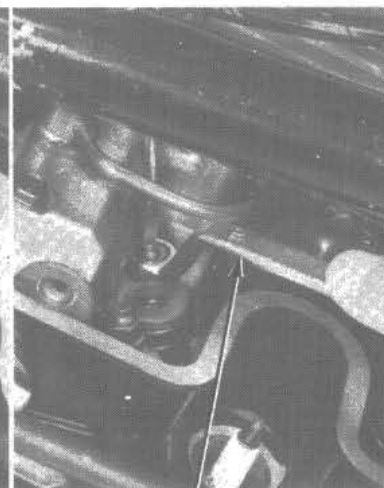


T1.3 MARK



REAR CRANKCASE MATING SURFACE

FEELER GAUGE



VALVE ADJUSTING WRENCH  
10 x 12 mm

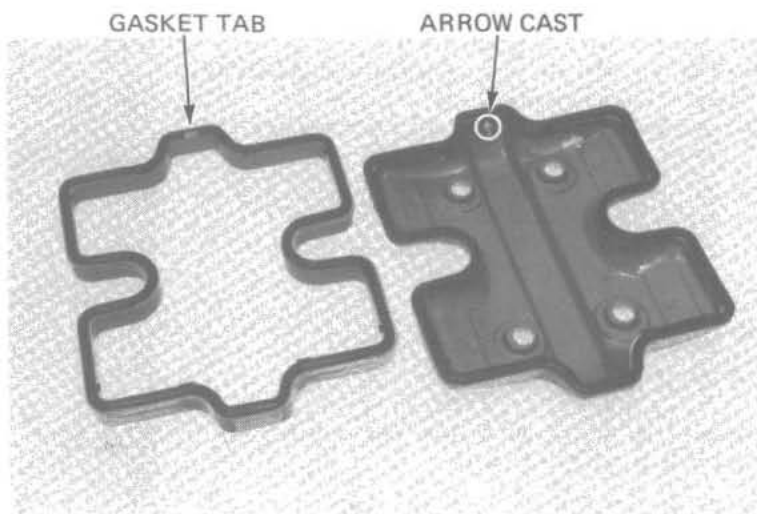




Install the removed parts in the reverse order of disassembly.

Be sure the cast marks inside the cylinder head cover face the front.

Also, the gasket tab of the rear cylinders head cover spacer should face forward.



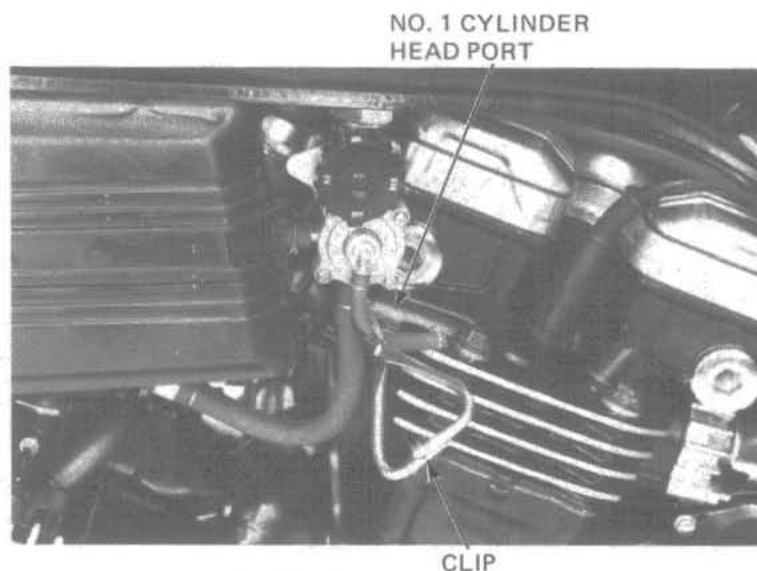
## CARBURETOR SYNCHRONIZATION

### NOTE

Synchronize the carburetors with the engine at normal operating temperature, transmission in neutral and motorcycle on the centerstand.

Remove the plugs from the No. 2, 3 and 4 cylinder head ports and install the vacuum gauge adapters.

Connect the vacuum gauges.



VACUUM GAUGE 07404-0020000  
or M937B-021-XXXXX (USA only)

### ADJUSTMENT

### NOTE

The No. 1 carburetor cannot be adjusted; It is the base carburetor.

Start the engine and adjust the idle speed.

**IDLE SPEED: 1,000 ± 100 rpm**

Clip the vacuum tube for the automatic fuel valve shut, as shown in the middle photo.

Stop the engine and remove the automatic fuel valve vacuum tube from the No. 1 cylinder head. Install the vacuum gauge adapter and connect the vacuum gauge.

Check that all carburetors are within 60 mm (2.4 in) Hg.





Synchronize to specification by loosening the adjusting screw lock nuts and turning the adjusting screws with the following tools:

To adjust the No. 2, 4 carburetor —

Carburetor adjusting wrench (07908-4220201)

To adjust the No. 3 carburetor —

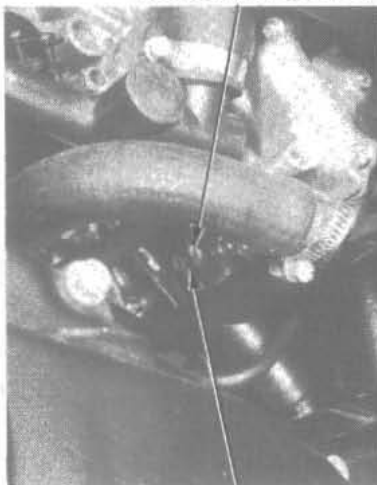
Carburetor throttle adjusting bit (07908-MB00000) or (KS-AH-008-MB0, USA only) and Adjusting wrench (07708-0030200) or Snap-On Tool Wrench X0810 or equivalent available in U.S.A.

Recheck the idle speed and synchronization.

Remove the gauge adapters and install the plugs.

Remove the clip from the automatic fuel valve vacuum tube. Reconnect the tube to the No. 1 cylinder.

ADJUSTING SCREW



CARBURETOR ADJUSTING  
WRENCH 07908-4220201



LOCK NUT

## CARBURETOR IDLE SPEED

### NOTE

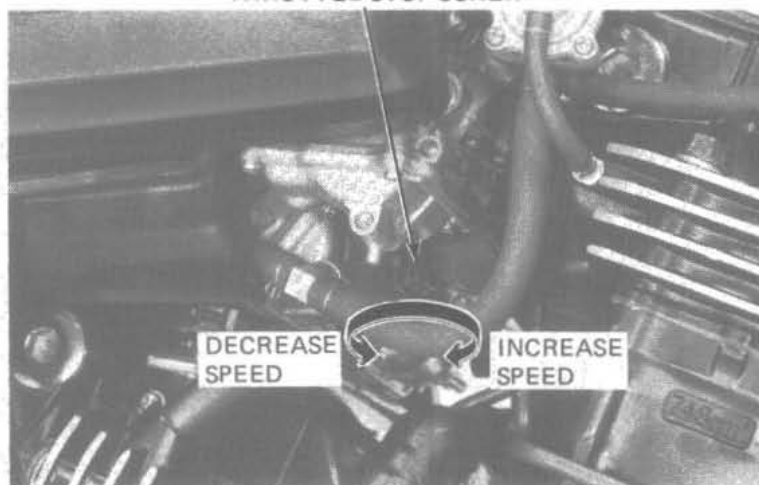
- Inspect and adjust idle speed after all other engine adjustments are within specifications.
- The engine must be warm for accurate adjustment. Ten minutes of stop-and-go riding is sufficient.

Warm up the engine, shift to NEUTRAL, and place the motorcycle on its center stand.

Turn the throttle stop screw as required to obtain the specified idle speed.

**IDLE SPEED: 1,000 ± 100 rpm**

THROTTLE STOP SCREW



## RADIATOR COOLANT

Remove the frame left side cover (V45 SABRE).

Remove the frame right side cover (V45 MAGNA).

Check the coolant level of the reserve tank with the engine running at normal operating temperature.

The level should be between the "FULL" and "LOW" level lines.

If necessary, remove the reserve tank cap and fill to the "FULL" level line with a 50/50 mixture of distilled water and anti-freeze.

Reinstall the cap and frame side cover.

RESERVE TANK CAP



"FULL" MARK

"LOW" MARK





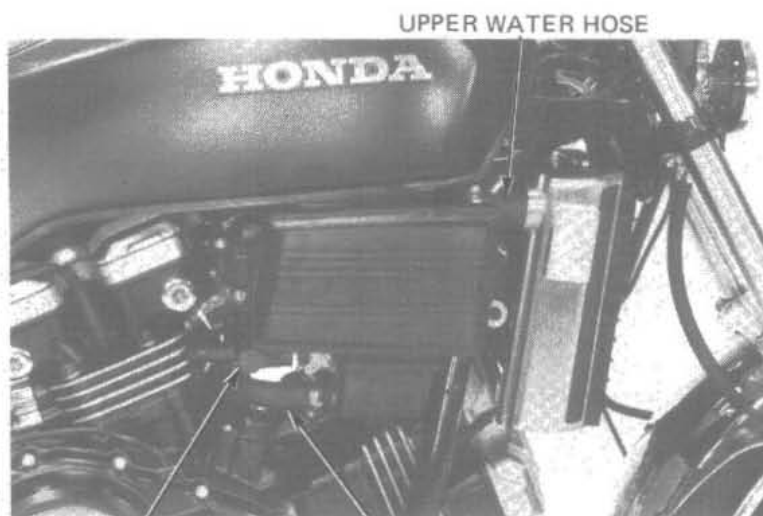
## RADIATOR CORE

Check the air passages for clogging or damage. Straighten bent fins, remove insects, mud or any obstruction with compressed air or low water pressure. Replace the radiator if the air flow is restricted over more than 20% of the radiating surface.



## COOLING SYSTEM HOSES

Inspect the hoses for cracks or deterioration, and replace if necessary. Check the tightness of all hose clamps.



## CYLINDER COMPRESSION

Warm up the engine. Stop the engine, then disconnect the spark plug caps and remove the spark plugs. Insert the compression gauge. Open the throttle all the way and crank the engine with the starter motor.

### NOTE:

Crank the engine until the gauge reading stops rising. The maximum reading is usually reached within 4-7 seconds.

### COMPRESSION PRESSURE:

$13 \pm 2 \text{ kg/cm}^2$  ( $184 \pm 28 \text{ psi}$ )

If compression is low, check for the following:

- Leaky valves
- Improper valve clearance
- Leaking cylinder head gasket
- Worn piston/ring/cylinder

If compression is high, it indicates that carbon deposits have accumulated on the combustion chamber and/or the piston crown.







## <CHASSIS>

### BATTERY

#### V45 SABRE:

Remove the right frame side cover and inspect the battery fluid level.

When the fluid level nears the lower level, remove the battery and add distilled water to the upper level line as follows:

Disconnect the negative cable at the battery terminal. Then remove the positive cable.

Remove the battery holder plate bolt and plate. Pull out the battery and add distilled water to the upper level line. Reinstall the battery.

#### V45 MAGNA:

Remove the right frame side cover and inspect the battery fluid level.

Remove the battery when the fluid level nears the lower level line as follows:

Remove the seat.

Disconnect the negative cable at the battery terminal. Then remove the positive cable.

Remove the battery holder plate bolt and plate. Pull out the battery and add distilled water to the upper level line.

Reinstall the battery.

#### NOTE:

Add only distilled water. Tap water will shorten the service life of the battery.

#### WARNING

*The battery electrolyte contains sulphuric acid. Protect your eyes, skin, and clothing. If electrolyte gets in your eyes, flush them thoroughly with water and get prompt medical attention.*

### BRAKE FLUID

Check the front brake fluid reservoir level.

If the level nears the lower level mark, fill the reservoir with DOT-3 BRAKE FLUID to the upper level mark.

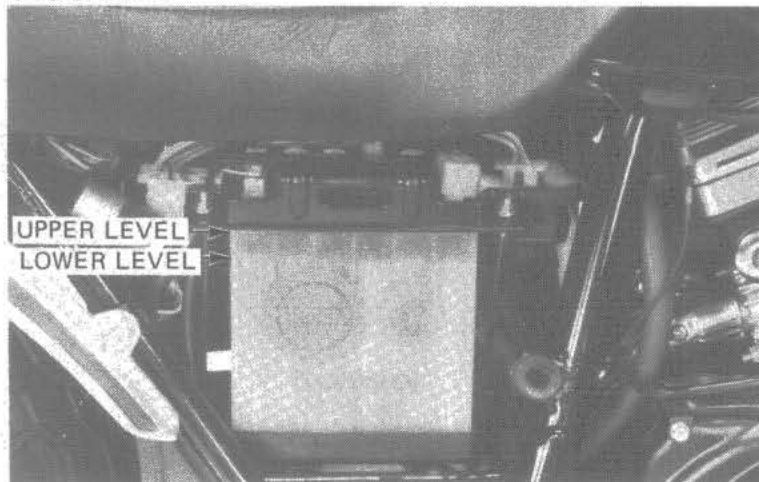
Check the entire system for leaks, if the level is low.

#### CAUTION:

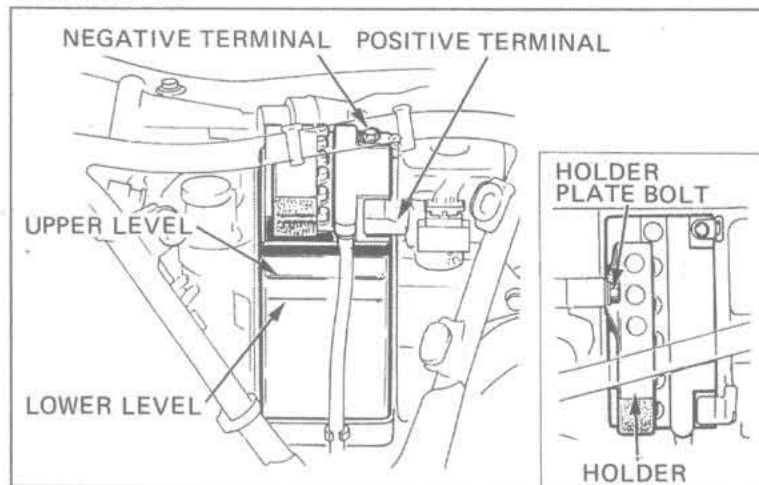
- Do not remove the cover until the handlebar has been turned so that the reservoir is level.
- Avoid operating the brake lever with the cap removed. Brake fluid will squirt out if the lever is pulled.
- Do not mix different types of fluid, as they are not compatible.

Refer to section 16 for brake bleeding procedures.

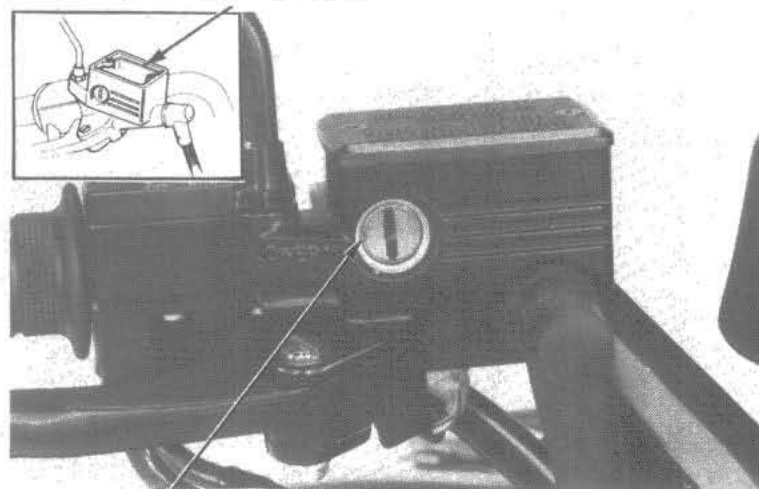
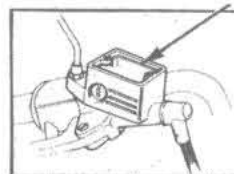
#### V45 SABRE:



#### V45 MAGNA:



#### UPPER LEVEL MARK



#### LOWER LEVEL MARK



## BRAKE SHOE/PAD WEAR

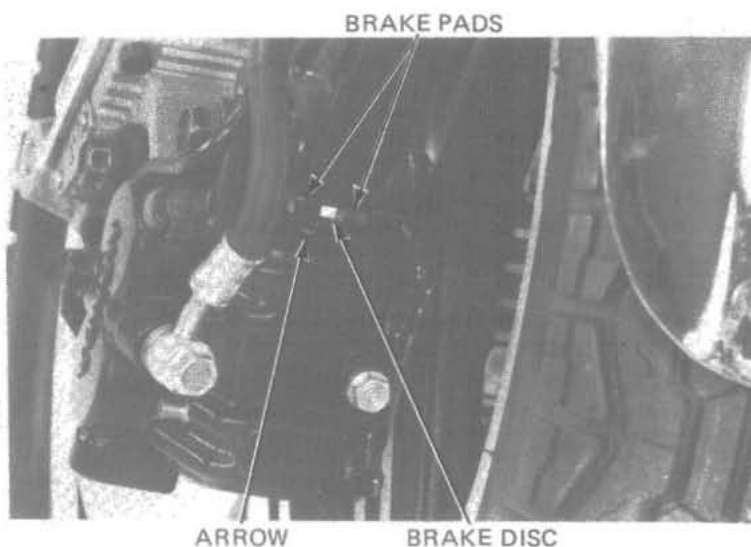
### BRAKE PAD WEAR

Check the brake pads for wear by looking through the slot indicated by the arrow cast on the caliper assembly.

Replace the brake pads if the wear line on the pads reaches the edge of the brake disc (page 16-5).

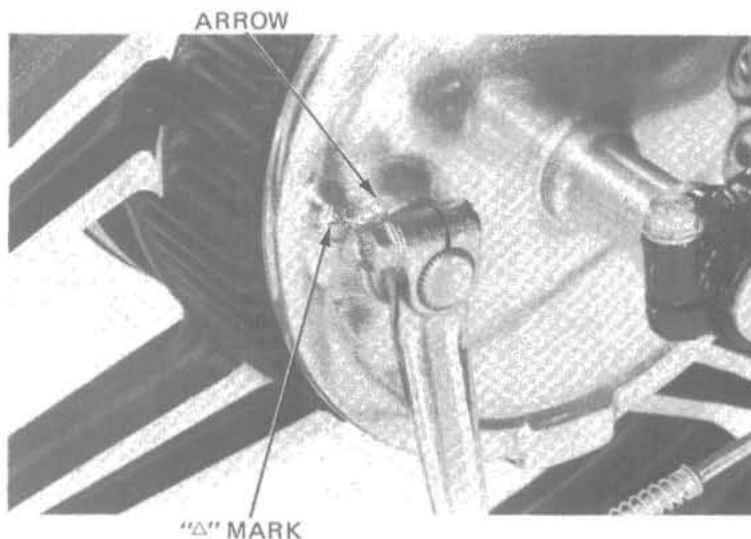
#### CAUTION:

*Always replace the brake pads in pairs to assure even disc pressure.*



### BRAKE SHOE INSPECTION (WEAR INDICATOR)

Replace the brake shoes if the arrow on the brake arm aligns with the reference mark "△" on full application of the rear brake pedal.



## BRAKE SYSTEM

Inspect the brake hoses and fittings for deterioration, cracks and signs of leakage. Tighten any loose fittings.

Replace hoses and fittings as required.

### BRAKE PEDAL HEIGHT

Adjust brake pedal height so the pedal is 7 mm (1/4 in) below the top of the foot peg.

#### CAUTION:

*Incorrect brake pedal height can cause brake drag.*

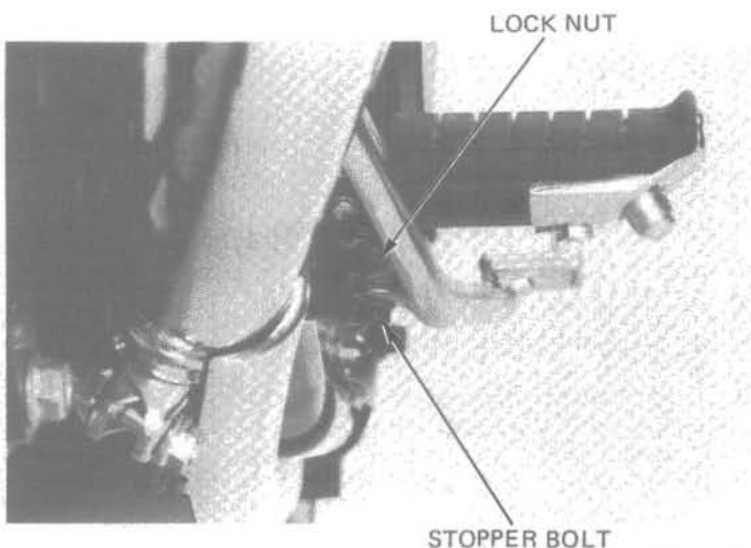
#### To Adjust:

Loosen the stopper bolt lock nut and turn the stopper bolt.

Retighten the lock nut.

#### NOTE:

After adjusting the brake pedal height, check the rear brake light switch and brake pedal free play and adjust if necessary.







## BRAKE PEDAL FREE PLAY

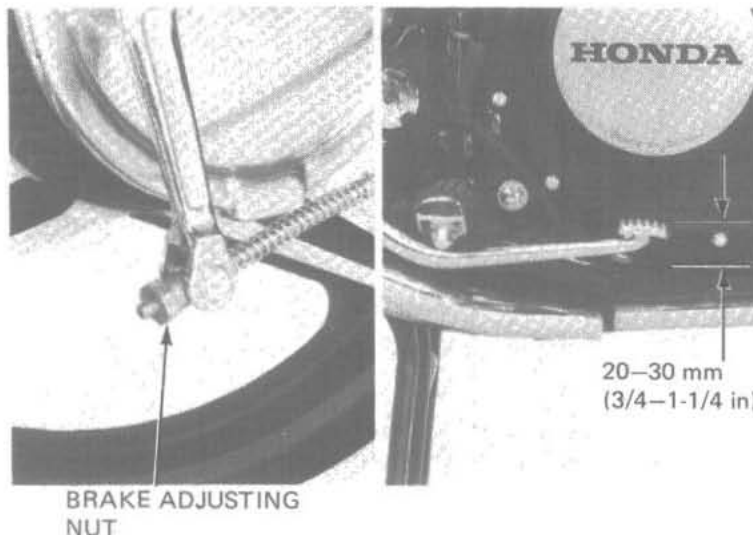
### NOTE:

Perform brake pedal free play adjustment after adjusting brake pedal height.

Check the brake pedal free play.

**FREE PLAY: 20–30 mm (3/4–1-1/4 in)**

If adjustment is necessary, turn the rear brake adjusting nut.



BRAKE ADJUSTING NUT

## BRAKE LIGHT SWITCH

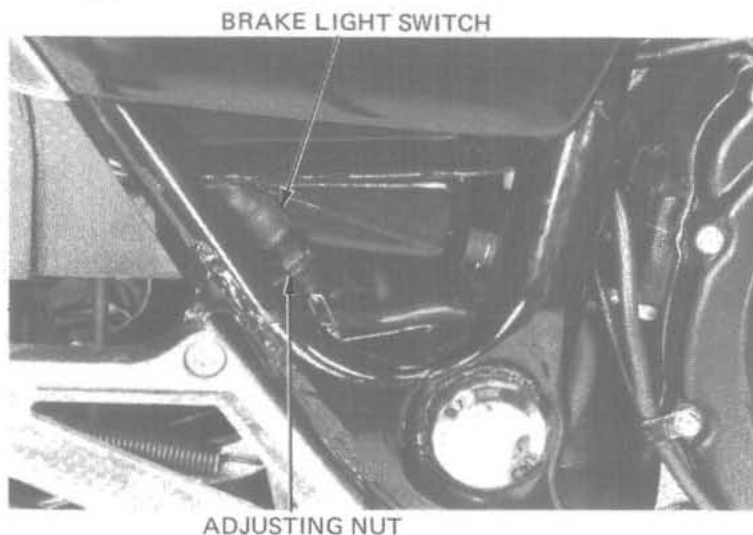
### NOTE:

Perform brake light switch adjustment after adjusting the brake pedal play and height.

Adjust the brake light switch so that the brake light will come on when the brake pedal is depressed 20 mm (3/4 in), and brake engagement begins. Adjust by holding the switch body and turning the adjusting nut. Do not turn the switch body.

### NOTE:

The front brake light switch does not require adjustment.



ADJUSTING NUT

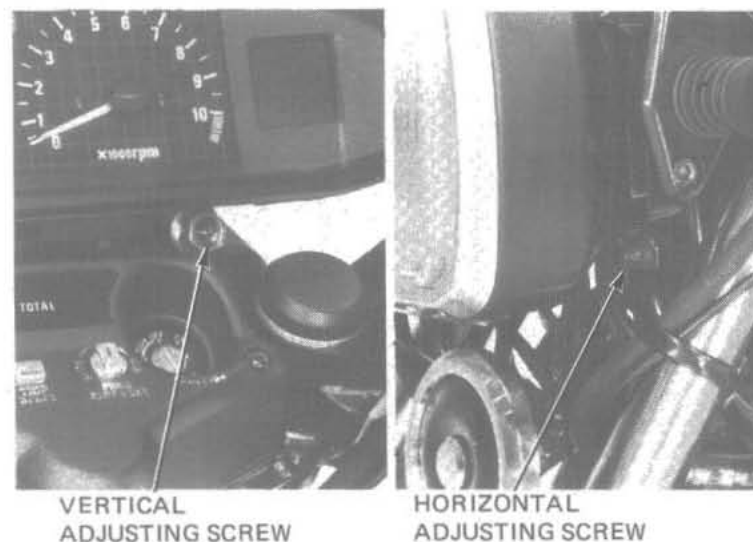
## HEADLIGHT AIM

### V45 SABRE:

Adjust vertically by turning the adjusting screw below the meter assembly (tachometer).

Adjust horizontally by turning the adjusting screw on the headlight case.

Turn the adjusting screw clockwise to direct the beam toward the right side of the rider.



VERTICAL ADJUSTING SCREW

HORIZONTAL ADJUSTING SCREW



## V45 MAGNA:

Adjust vertically by loosening both headlight case mounting bolts.

Adjust horizontally by turning the adjusting screw on the headlight rim.

Turn the adjusting screw clockwise to direct the beam toward the right side of the rider.

### NOTE:

Adjust the headlight beam as specified by local laws and regulations.

### WARNING

*An improperly adjusted headlight may blind oncoming drivers, or it may fail to light the road for a safe distance.*

## CLUTCH

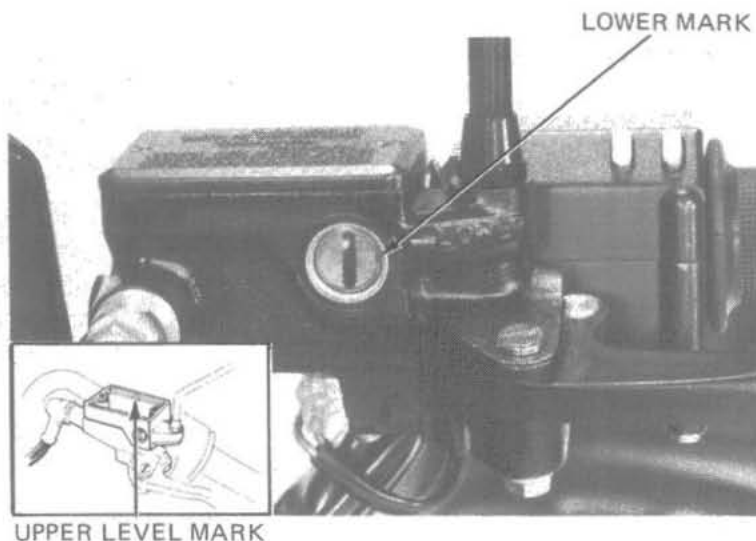
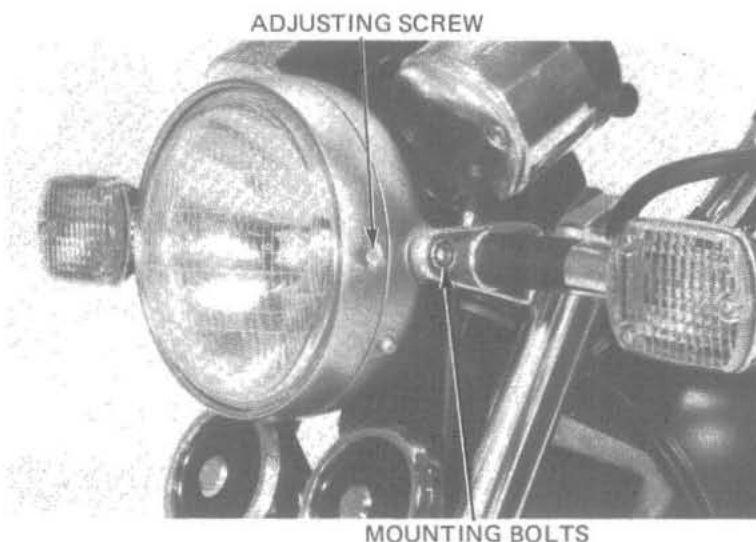
Check the clutch fluid reservoir level.

If the level nears the lower level mark, fill the reservoir with DOT-3 BRAKE FLUID to the upper level mark.

Check the entire system for leaks, if the level is low.

### CAUTION:

- Do not remove the cover until the handlebar has been turned so that the reservoir is level.
- Avoid operating the clutch lever with the cap removed. Fluid will squirt out if the lever is pulled.
- Do not mix different types of fluid, as they are not compatible.

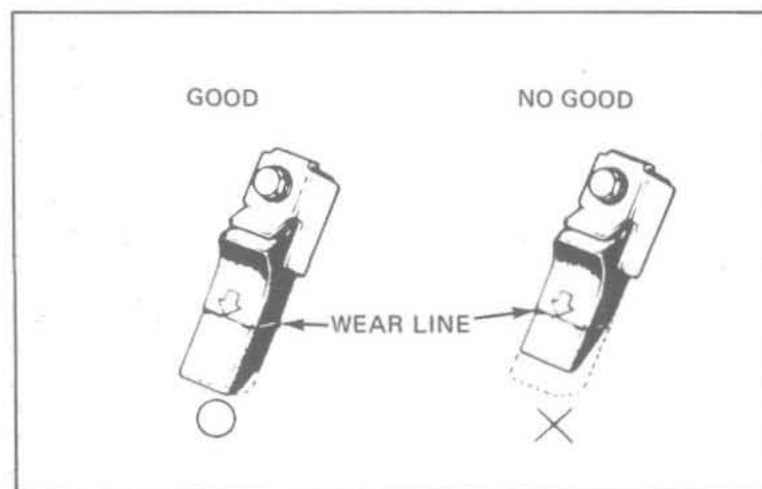


## SIDE STAND

Check the rubber pad for deterioration or wear. Replace if any wear extends to wear line as shown. Check the side stand spring for damage and loss of tension, and the side stand assembly for freedom of movement. Make sure the side stand is not bent.

### NOTE:

- When replacing, use a rubber pad with the mark "Over 260 lbs ONLY".
- Spring tension is correct if the measurements fall within 2–3 kg (4.4–6.6 lb), when pulling the side stand lower end with a spring scale.







## SUSPENSION

### WARNING

*Do not ride a vehicle with faulty suspension. Loose, worn or damaged suspension parts impair vehicle stability and control.*

### FRONT

Check the action of the front forks by compressing them several times.

Check the entire fork assembly for leaks or damage. Replace damaged components which cannot be repaired.

Tighten all nuts and bolts.



Check the front fork air pressure when the forks are cold.

Place the vehicle on its center stand.

Remove each air valve cap and measure the air pressure.

### AIR PRESSURE:

6–14 psi (40–100 kPa, 0.4–1.0 kg/cm<sup>2</sup>)



### ANTI-DIVE SYSTEM INSPECTION

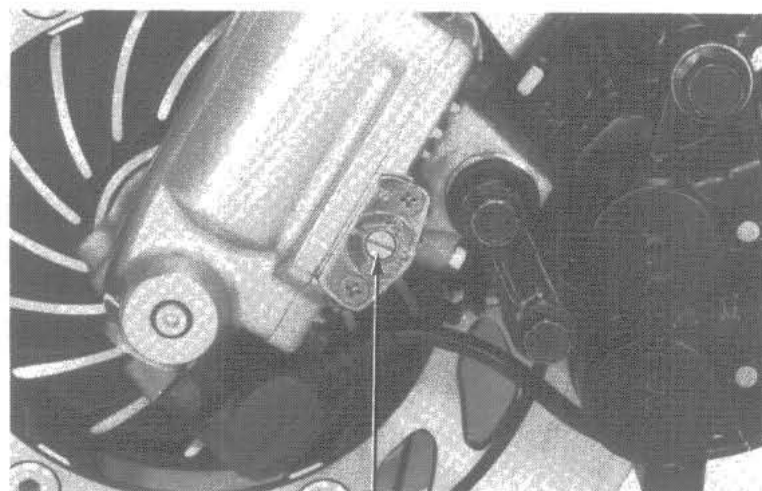
### WARNING

*Select a safe place away from traffic to perform this inspection.*

Check the operation of the anti-dive system by riding the motorcycle and firmly applying the brakes.

| Position | Anti-dive damper force |
|----------|------------------------|
| I        | LIGHT ANTI-DIVE        |
| II       | MEDIUM                 |
| III      | HARD                   |
| IV       | MAXIMUM ANTI-DIVE      |

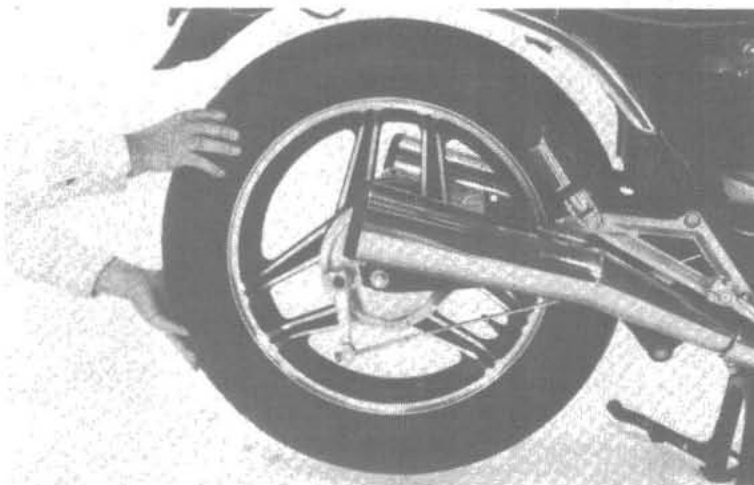
Inspect and if necessary, repair the system (Refer to section 15).





## REAR

Place the motorcycle on its center stand.  
 Move the rear wheel sideways with force to see if the swingarm bearings are worn.  
 Replace the bearings if there is any looseness (page 18-4).  
 Check the shock absorbers for leaks or damage.  
 Tighten all rear suspension nuts and bolts.



## V45 SABRE:

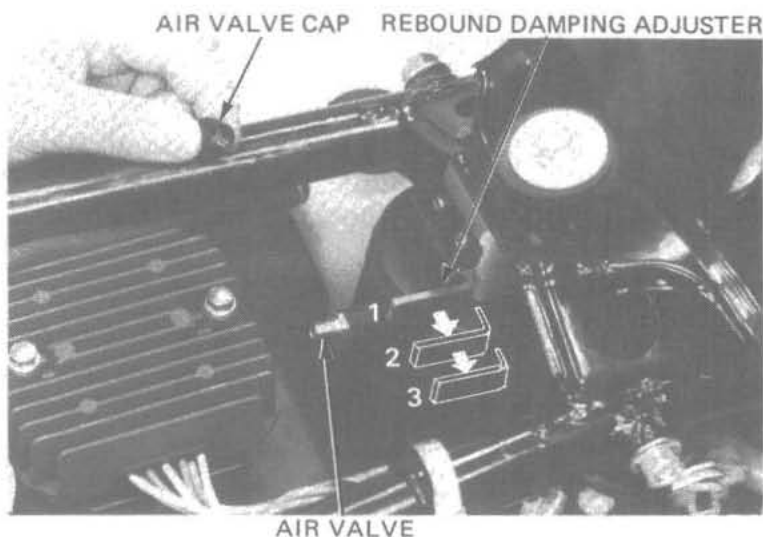
Remove the seat.  
 Remove the valve cap and measure the rear shock absorber air pressure.

**REAR SHOCK ABSORBER AIR PRESSURE:**  
**14–57 psi (100–400 kPa, 1–4 kg/cm<sup>2</sup>)**

### NOTE:

Check the air pressure when the rear shock absorber is cold.

Adjust the rebound damping according to the following chart.



Recommended Rear Suspension Adjustment:

| Rear Air Pressure                                    | Rider/Load                       | REBOUND DAMPING ADJUSTER | RIDING CONDITIONS              |
|------------------------------------------------------|----------------------------------|--------------------------|--------------------------------|
| 14 psi<br>(100 kPa,<br>1.0 kg/cm <sup>2</sup> )      | One<br>↑                         | 1                        | General or around town riding. |
| ↓<br>57 psi<br>(400 kPa,<br>4.0 kg/cm <sup>2</sup> ) | ↓<br>Up to vehicle capacity load | 2                        | Highway or winding road riding |
|                                                      |                                  | 3                        | Rough road riding              |





## WHEELS

### NOTE:

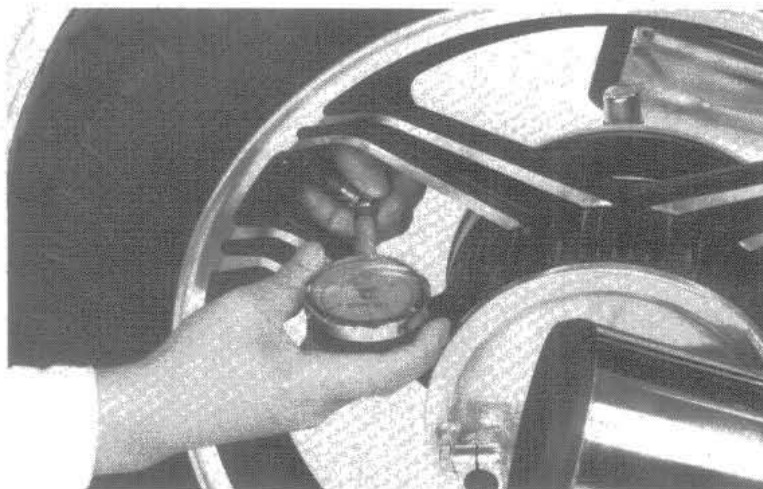
Tire pressure should be checked when tires are COLD.

Check the tires for cuts, imbedded nails, or other sharp objects.

### RECOMMENDED TIRES AND PRESSURES:

[ ] -- V45 MAGNA

|                                                   |                                               | Front             | Rear                             |
|---------------------------------------------------|-----------------------------------------------|-------------------|----------------------------------|
| Tire size                                         |                                               | 110/90-18 61H     | 130/90-17 68H<br>[130/90-16 67H] |
| Cold tire pressure psi (kPa, kg/cm <sup>2</sup> ) | Up to 90 kg (200 lbs) load                    | 32<br>(225, 2.25) | 32<br>(225, 2.25)                |
|                                                   | 90 kg (200 lbs) load to vehicle capacity load | 32<br>(225, 2.25) | 40<br>(280, 2.8)                 |
| Tire brand                                        | BRIDGE-STONE                                  | L303              | G508                             |
|                                                   | DUNLOP                                        | F11               | K127C                            |



Check the front and rear wheels for trueness (page 15-21 and 17-5).

Measure the tread depth at the center of the tires. Replace the tires if the tread depth reaches the following limit:

### Minimum tread depth:

Front: 1.5 mm (1/16 in)

Rear: 2.0 mm (3/32 in)

## STEERING HEAD BEARINGS

### NOTE:

Check that the control cables do not interfere with handlebar rotation.

Raise the front wheel off the ground and check that the handlebar rotates freely.

If the handlebar moves unevenly, binds, or has vertical movement, adjust the steering head bearing by turning the steering head adjusting nut (page 16-38).

## NUTS, BOLTS, FASTENERS

Check that all chassis nuts and bolts are tightened to their correct torque values (Section 1) at the intervals shown in the Maintenance Schedule (Pages 3-3 and 3-4).

Check all cotter pins, safety clips, hose clamps and cable stays.



